

Conditional Expectation

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Overview

Great Expectations (18.4)

- Alternate Definition of Expectation (18.4.4)

- Conditional Expectation (18.4.5)

- Mean Time to Failure (18.4.6)

- Expected Returns in Gambling Games (18.4.7)

Shortcut formula for expectation

Instead of summing over all outcomes, sometimes easier to group outcomes by the blocks defined by the random variable.

$$\mathbb{E}[R] = \sum_{x \in \text{range}(R)} x \cdot \Pr[R = x].$$

Proof:

$$\begin{aligned}
 \mathbb{E}[R] &= \sum_{\omega \in \mathcal{S}} R(\omega) \Pr[\omega] && \text{defn expt} \\
 &= \sum_{x \in \text{range}(R)} \sum_{\omega \in [R=x]} R(\omega) \Pr[\omega] && \text{range is partition} \\
 &= \sum_{x \in \text{range}(R)} \sum_{\omega \in [R=x]} x \Pr[\omega] && \text{defn } R, \omega \\
 &= \sum_{x \in \text{range}(R)} x \sum_{\omega \in [R=x]} \Pr[\omega] && \text{factor out } x \\
 &= \sum_{x \in \text{range}(R)} x \Pr[R = x]. && \text{defn event prob}
 \end{aligned}$$

Median

Not the same as mean or expected value. It's the probability-weighted middle of the set of values.

Definition: The *median* of a random variable R is a value $x \in \text{range}(R)$ such that

$$\Pr[R \leq x] \geq \frac{1}{2}$$

and

$$\Pr[R \geq x] \geq \frac{1}{2}.$$

Book: “We won’t devote much attention to the median. The expected value is more useful and has much more interesting properties.”

Me: Ptth.

Also: The book’s definition is wrong.

Definition of conditional expectation

Definition: The conditional expectation of a random variable R given event A is:

$$\mathbb{E}[R|A] := \sum_{r \in \text{range}(R)} r \cdot \Pr[R = r|A].$$

Example: R is the deck of cards example from before: value of card, 1 for ace, 10 for face.

$$\begin{aligned} \mathbb{E}[R|R < 10] &= \sum_{i=1}^9 i \Pr[R = i|R < 10] \\ &= \sum_{i=1}^9 i \frac{\Pr[R=i \text{ AND } R < 10]}{\Pr[R < 10]} \\ &= \sum_{i=1}^9 i \frac{1/13}{9/13} \\ &= 1/9 \sum_{i=1}^9 i \\ &= 45/9 = 5. \end{aligned}$$

The height of usefulness

How tall are elephants on average? Don't know. Web search didn't help. Conditional expectations can!

$$\begin{aligned}\mathbb{E}[H] &= \mathbb{E}[H|Af] \cdot \Pr[Af] + \mathbb{E}[H|As] \cdot \Pr[As] && \text{split} \\ &= 10 \cdot 0.9 + \mathbb{E}[H|As] \cdot 0.1 && \text{Google} \\ &= 10 \cdot 0.9 + (\mathbb{E}[H|As \text{ AND } m] \Pr[m|As] \\ &\quad + \mathbb{E}[H|As \text{ AND } f] \cdot \Pr[f|As]) \cdot 0.1 && \text{split} \\ &= 10 \cdot 0.9 + (9 \cdot 1/2 + 7 \cdot 1/2) \cdot 0.1 && \text{Google} \\ &= 9.8 && \text{calculator}\end{aligned}$$

Memoryless crash time

A computer program crashes at the end of each hour of use with probability p , if it has not crashed already. Let C be the random variable for the time until a crash occurs. What is $\mathbb{E}[C]$?

Let A be the event that the program crashes after the first hour and $\bar{A}$ be the complementary event. Mean time to failure is

$$\mathbb{E}[C] = \mathbb{E}[C|A] \cdot \Pr[A] + \mathbb{E}[C|\bar{A}] \cdot \Pr[\bar{A}]$$

$\mathbb{E}[C|A] = 1$ because A is the event that the program crashes after one hour.

$\mathbb{E}[C|\bar{A}] = 1 + \mathbb{E}[C]$ because the program runs for an hour, then we're back in the original situation.

$$\mathbb{E}[C] = 1 \cdot p + (1 + \mathbb{E}[C])(1 - p)$$

$$\mathbb{E}[C] = p + 1 + \mathbb{E}[C] - p - p\mathbb{E}[C]$$

$$p\mathbb{E}[C] = 1$$

$$\mathbb{E}[C] = 1/p$$

Mean babies until boy

If a couple decides to keep having babies until one is a boy, assuming there are just boys and girls and they are equally likely, how many kids will they make?

failure hour p $1/p$	boy baby $1/2$ 2
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- ▶ Coin flips until heads? 2
- ▶ Cards until face card? $13/3 = 4.3$ ish
- ▶ Kobe (0.85) free throws until basket? 1.18

Expected winnings

D dollars won, w event that you win, $\bar{w}$ event that you lose.

$$\mathbb{E}[D] = \mathbb{E}[D|w] \Pr[w] + \mathbb{E}[D|\bar{w}](1 - \Pr[w])$$

Game is *fair* if $\mathbb{E}[D] = 0$.

Example: $\mathbb{E}[D|w] = -\mathbb{E}[D|\bar{w}]$, $\Pr[w] = 1/2$.

Example: SAT points (before March 2016). $\mathbb{E}[D|w] = 1$,
 $\mathbb{E}[D|\bar{w}] = -1/4$, $\Pr[w] = 1/5$.

$$1 \cdot .2 + (-1/4) \cdot .8 = 0.$$

Bar bet

Three people. Each player puts in \$2. Each player predicts the outcome of a coin flip. Correct guessers split the pot. If all wrong, money returned.

$$\begin{aligned}\mathbb{E}[D] &= \mathbb{E}[D|\text{win alone}] \Pr[\text{win alone}] \\ &\quad + \mathbb{E}[D|\text{win with one}] \Pr[\text{win with one}] \\ &\quad + \mathbb{E}[D|\text{win with both}] \Pr[\text{win with both}] \\ &\quad + \mathbb{E}[D|\text{lose}] \Pr[\text{lose}] \\ &\quad + \mathbb{E}[D|\text{all lose}] \Pr[\text{all lose}]\end{aligned}$$

$$\begin{aligned}\mathbb{E}[D] &= 4 \cdot \Pr[\text{win alone}] \\ &\quad + 1 \cdot \Pr[\text{win with one}] \\ &\quad + 0 \cdot \Pr[\text{win with both}] \\ &\quad - 2 \cdot \Pr[\text{lose}] \\ &\quad + 0 \cdot \Pr[\text{all lose}]\end{aligned}$$

Random guesses

$$\begin{aligned}\mathbb{E}[D] &= 4 \cdot \Pr[\text{win alone}] \\ &\quad + 1 \cdot \Pr[\text{win with one}] \\ &\quad + 0 \cdot \Pr[\text{win with both}] \\ &\quad - 2 \cdot \Pr[\text{lose}] \\ &\quad + 0 \cdot \Pr[\text{all lose}]\end{aligned}$$

Everyone guesses randomly.

- ▶ $\Pr[\text{win alone}] = 1/8$
- ▶ $\Pr[\text{win with one}] = 1/4$
- ▶ $\Pr[\text{win with both}] = 1/8$
- ▶ $\Pr[\text{lose}] = 3/8$
- ▶ $\Pr[\text{all lose}] = 1/8$

$$4 \cdot 1/8 + 1 \cdot 1/4 + 0 \cdot 1/8 - 2 \cdot 3/8 + 0 \cdot 1/8 = 0. \text{ Fair!}$$

Collusion

$$\begin{aligned}\mathbb{E}[D] &= 4 \cdot \Pr[\text{win alone}] \\ &\quad + 1 \cdot \Pr[\text{win with one}] \\ &\quad + 0 \cdot \Pr[\text{win with both}] \\ &\quad - 2 \cdot \Pr[\text{lose}] \\ &\quad + 0 \cdot \Pr[\text{all lose}]\end{aligned}$$

The other two collude to make opposite guesses.

- ▶ $\Pr[\text{win alone}] = 0$
- ▶ $\Pr[\text{win with one}] = 1/2$
- ▶ $\Pr[\text{win with both}] = 0$
- ▶ $\Pr[\text{lose}] = 1/2$
- ▶ $\Pr[\text{all lose}] = 0$

$$4 \cdot 0 + 1 \cdot 1/2 + 0 \cdot 0 - 2 \cdot 1/2 + 0 \cdot 0 = -0.5. \text{ I lose!}$$